

0.1 Review for Exam 3

0.1.1 Wilson's Theorem

Theorem 1 (Wilson's Theorem)

If p is a prime integer, then

$$(p-1)! \equiv p-1 \pmod{p}$$

or

$$(p-1)! \equiv -1 \pmod{p}$$

But for composite numbers, we have

Theorem 2 (Some Unnamed Theorem)

If n is a composite integer greater than 4,

then

$$(n-1)! \equiv 0 \pmod{n}$$

If $n = 4$, then

$$3! \equiv 2 \pmod{4}$$

0.1.2 Fermat's Little Theorem

Theorem 3 (Fermat's Little Theorem)

If $a \in \mathbb{Z}$ with,

1. p is prime
2. $\gcd(a, p) = 1$

$$a^{p-1} \equiv 1 \pmod{p}$$

0.1.3 Extended Fermat's Little Theorem

Theorem 4 (Extended Fermat's Little Theorem)

If $a, p, q \in \mathbb{Z}$ with

1. p and q are distinct primes

$$2. \gcd(a, pq) = 1$$

Then

$$a^{(p-1)(q-1)} \equiv 1 \pmod{pq}$$

0.1.4 Bezout's Theorem

Theorem 5 (Bézout's Theorem)

Basically

$$a, b \in \mathbb{Z}^+ \implies \exists x, y \in \mathbb{Z}, ax + by = \gcd(a, b)$$

When given a problem that's like find x, y in $ax + by = \gcd(a, b)$ (where they give you a and b), just do Euclidean and stop when you get to the actual GCD then back solve

0.1.5 Is a number prime or composite?

Lowkey just brute force up until

$$\lceil \sqrt{n} \rceil$$

0.1.6 Prime Factorization

Let's look at how to find the number of divisors in a number n

Definition 1 (Prime Factorization)

Prime factorization is breaking up a number into prime factors. An example is

$$\begin{aligned} 24 &= 6 \cdot 4 \\ &= (3 \cdot 2)(2 \cdot 2) \\ &= 3^1 \cdot 2^3 \end{aligned}$$

The prime factors are 3 and 2.

In order to find the number of divisors, we must first break up n into its prime factors

$$n = p_1^{e_1} \cdot p_2^{e_2} \cdot \dots \cdot p_k^{e_k}$$

p_k : Prime factor

e_k : Occurences

The number of divisors is then

$$(e_1 + 1)(e_2 + 1) \dots (e_k + 1)$$

0.1.7 Chinese Remainder Theorem

Given the system: (I'm just gonna use up to 3 for this)

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

$$x \equiv a_3 \pmod{m_3}$$

The system has a **unique solution** if m_1, m_2, m_3 are pairwise coprime. If not, then it *could possibly* have a solution but not necessarily

1. Expand the first equation

$$x \equiv a_1 \pmod{m_1}$$

implies

$$\boxed{x = m_1 k_1 + a_1}, k_1 \in \mathbb{Z}$$

2. Plug that into the second equation

$$x \equiv a_2 \pmod{m_2}$$

$$m_1 k_1 + a_1 \equiv a_2 \pmod{m_2}$$

And solve for k_1

$$\begin{aligned}
m_1 k_1 + a_1 &\equiv a_2 \pmod{m_2} \\
m_1 k_1 &\equiv a_2 - a_1 \pmod{m_2} \\
m_1^{-1} k_1 &\equiv m_1^{-1}(a_2 - a_1) \pmod{m_2} \\
k_1 &\equiv m_1^{-1}(a_2 - a_1) \pmod{m_2} \\
\boxed{k_1 \equiv c_1} &\pmod{m_2} \quad (c_1 \text{ is shorthand})
\end{aligned}$$

Note

This isn't a formula, just steps with arbitrary variables. You'll need to find the inverse of m_1

And expand that last equation

$$k_1 \equiv c_1 \pmod{m_2}$$

implies

$$\boxed{k_1 = m_2 k_2 + c_1}, k_2 \in \mathbb{Z}$$

3. Plug that into our equation for x

$$\begin{aligned}
x &= m_1 k_1 + a_1 \\
\boxed{x &= m_1(m_2 k_2 + c_1) + a_1}
\end{aligned}$$

Plug that into the last equation

$$m_1(m_2 k_2 + c_1) + a_1 \equiv a_3 \pmod{m_3}$$

and solve for $k_2 \pmod{m_3}$

$$\vdots$$

$$\boxed{k_2 \equiv c_2 \pmod{m_3}}$$

Expand that equation

$$\boxed{k_2 = m_3 k_3 + c_2}, k_3 \in \mathbb{Z}$$

4. Finally, plug that last equation into our SECOND equation for x

$$x = m_1(m_2 k_2 + c_1) + a_1$$

$$x = m_1(m_2(m_3 k_3 + c_2) + c_1) + a_1$$

$$\boxed{x = m_1 m_2 m_3 k_3 + c_3}$$

Now we UN-expand this equation and see that

$$\boxed{x \equiv c_3 \pmod{m_1 m_2 m_3}}$$

Basically, starting with the first equation

1. Expand first, solving for x
2. Plug x into second
3. Solve for k_1
4. Expand k_1
5. Plug k_1 back into first expanded of x (one with k_1)
6. Solve for k_2
7. Expand k_2
8. Plug k_2 back into second expanded of x (one with k_2)
9. Unexpand back into modulus

Just do these lines for the first equation, and these lines for each other equation